

CLASS_DIAGRAM

Class Diagrams

Package overview

```
classDiagram
    direction TB
    class AuthenticationController
    class AuthenticationService
    class JwtService
    class JwtAuthenticationFilter
    class CustomUserDetailsService
    class BankUserDetails

    AuthenticationController --> AuthenticationService
    AuthenticationService --> JwtService
    AuthenticationService --> CustomUserDetailsService
    JwtAuthenticationFilter --> JwtService
    JwtAuthenticationFilter --> CustomUserDetailsService
    CustomUserDetailsService --> BankUserDetails
```

Account domain

```
classDiagram
    direction TB
    class AccountController {
        +openAccount()
        +searchAccounts()
        +getAccountsByCustomer()
        +updateAccountStatus()
        +deposit()
    }
    class AccountService
    class AccountServiceImpl {
        +openAccount()
        +deposit()
        +searchAccounts()
        +updateAccountStatus()
    }
    class AccountRepository
    class Account
    class AccountNumberGenerator
    class AccountSpecification
    class AccountPolicyController {
        +createPolicy()
        +getActivePolicy()
    }
    class AccountPolicyService
    class AccountPolicyServiceImpl
    class AccountPolicyRepository
    class AccountPolicy
    class Customer

    AccountController --> AccountService
    AccountService <|.. AccountServiceImpl
    AccountServiceImpl --> AccountRepository
```

```

AccountServiceImpl --> AccountPolicyRepository
AccountServiceImpl --> AccountNumberGenerator
AccountServiceImpl --> Customer
AccountRepository --> Account
AccountSpecification ..> Account
Account --> Customer : customer_id
AccountPolicyController --> AccountPolicyService
AccountPolicyService <|.. AccountPolicyServiceImpl
AccountPolicyServiceImpl --> AccountPolicyRepository
AccountPolicyRepository --> AccountPolicy

```

Customer domain

```

classDiagram
    direction TB
    class CustomerController
    class CustomerService
    class CustomerServiceImpl {
        +createCustomer()
        +updateCustomer()
        +searchCustomers()
        +deleteCustomer()
    }
    class CustomerRepository
    class Customer
    class AccountService

    CustomerController --> CustomerService
    CustomerService <|.. CustomerServiceImpl
    CustomerServiceImpl --> CustomerRepository
    CustomerServiceImpl --> AccountService
    CustomerRepository --> Customer

```

User / role domain

```

classDiagram
    direction TB
    class UserController
    class UserService
    class User
    class Role
    class UserRole
    class UserRepository
    class RoleRepository
    class UserRoleRepository

    UserController --> UserService
    UserService --> UserRepository
    UserService --> UserRoleRepository
    UserService --> RoleRepository
    User --> UserRole
    Role --> UserRole

```

Shared

```

classDiagram
    direction TB
    class AuditableEntity {
        createdAt
        updatedAt
    }

```

```
    createdBy
    updatedBy
    version
}
class AccountPolicy
class User
class Role
class UserRole
class SecurityConfig
class GlobalExceptionHandler

AuditableEntity <|-- AccountPolicy
AuditableEntity <|-- User
AuditableEntity <|-- Role
AuditableEntity <|-- UserRole
```

Frontend feature map (logical)

```
flowchart LR
    Login --> AuthService
    Dashboard --> DashboardService
    Customers --> CustomerService
    Accounts --> AccountService
    OpeningDeposit --> AccountPolicyService
    Employees --> UserService
    AuthService --> API
    CustomerService --> API
    AccountService --> API
    AccountPolicyService --> API
    UserService --> API
    DashboardService --> API
```